TESTING UNIVERSAL CLOSEDNESS ON AFFINE SPACE FIBRES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4643-4732, following the statement at 4628-4641.

Editorial note. The '(some details omitted)' covers the etale-local-on-Z reduction of closedness of the topological map; flagged in the notes.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 4699 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Separation lemma for non-closed images. lemma-separate (same chapter, cited at ~4650-4651). Produces the etale neighbourhood (V,v), the diagram through Z', and T'.
- Points of cartesian squares. Properties of Spaces, Lemma spaces-properties-lemma-points-cartesian (cross-chapter, cited at ~4679-4680). Identifies the image of T_V in |V| with the pullback of the image of T.
- Universal closedness is etale local on the target. Morphisms of Spaces, Lemma spaces-morphisms-lemma-universally-closed-local (cross-chapter, cited at ~4690-4692). Lifts universal closedness of X x_Y V -> V back to f.
- Quasi-affine immersion into affine space over the base. Morphisms, Lemma morphisms-lemma-quasi-affine-finite-type-over-S (cross-chapter, cited at ~4701-4702). Immerses Z into A^n x Y.
- Subspace topology on topological images. Properties of Spaces, Lemma spaces-properties-lemma-subspace-induced-topology (cross-chapter, cited at ~4725-4726). Horizontal arrows are homeomorphisms onto locally closed subsets.

Source reference. Original lemma 0CM8: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 4529–4529. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04II: focused-seeds/geometry/S9-spaces-limits-test-universally-closed/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 03IT: focused-seeds/geometry/S9-spaces-limits-test-universally-closed/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03H4: focused-seeds/geometry/S9-spaces-limits-test-universally-closed/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0ABJ: focused-seeds/geometry/S9-spaces-limits-test-universally-closed/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a quasi-compact morphism of algebraic spaces over S. The following are equivalent*

(1) *f is universally closed,*
(2) *for every morphism $Z \to Y$ which is locally of finite presentation the map $|X \times_Y Z| \to |Z|$ is closed, and*
(3) *there exists a scheme V and a surjective étale morphism $V \to Y$ such that $|\mathbf{A}^n \times (X \times_Y V)| \to |\mathbf{A}^n \times V|$ is closed for all $n \geq 0$.*

Proof. It is clear that (1) implies (2). Suppose that $|X \times_Y Z| \to |Z|$ is not closed for some morphism of algebraic spaces $Z \to Y$ over S. This means that there exists some closed subset $T \subset |X \times_Y Z|$ such that $\mathrm{Im}(T \to |Z|)$ is not closed. Pick $z \in |Z|$ in the closure of the image of T but not in the image. Apply Lemma 0CM8. We find an étale neighbourhood $(V, v) \to (Z, z)$, a commutative diagram

$$\begin{array}{ccc} V & \xrightarrow{a} & Z' \\ \downarrow & & \downarrow b \\ Z & \xrightarrow{g} & Y, \end{array}$$

and a closed subset $T' \subset |X \times_Y Z'|$ such that

(1) the morphism $b : Z' \to Y$ is locally of finite presentation,
(2) with $z' = a(v)$ we have $z' \notin \mathrm{Im}(T' \to |Z'|)$, and
(3) the inverse image of T in $|X \times_Y V|$ maps into T' via $|X \times_Y V| \to |X \times_Y Z'|$.

We claim that z' is in the closure of $\mathrm{Im}(T' \to |Z'|)$ which implies that $|X \times_Y Z'| \to |Z'|$ is not closed. The claim shows that (2) implies (1). To see the claim is true we contemplate following commutative diagram

$$\begin{array}{ccccc} X \times_Y Z & \longleftarrow & X \times_Y V & \longrightarrow & X \times_Y Z' \\ \downarrow & & \downarrow & & \downarrow \\ Z & \longleftarrow & V & \xrightarrow{a} & Z' \end{array}$$

Let $T_V \subset |X \times_Y V|$ be the inverse image of T. By Properties of Spaces, Lemma 03H4 the image of T_V in $|V|$ is the inverse image of the image of T in $|Z|$. Then

since z is in the closure of the image of $T \to |Z|$ and since $|V| \to |Z|$ is open, we see that v is in the closure of the image of $T_V \to |V|$. Since the image of T_V in $|X \times_Y Z'|$ is contained in $|T'|$ it follows immediately that $z' = a(v)$ is in the closure of the image of T'.

It is clear that (1) implies (3). Let $V \to Y$ be as in (3). If we can show that $X \times_Y V \to V$ is universally closed, then f is universally closed by Morphisms of Spaces, Lemma 03IT. Thus it suffices to show that $f : X \to Y$ satisfies (2) if f is a quasi-compact morphism of algebraic spaces, Y is a scheme, and $|\mathbf{A}^n \times X| \to |\mathbf{A}^n \times Y|$ is closed for all n. Let $Z \to Y$ be locally of finite presentation. We have to show the map $|X \times_Y Z| \to |Z|$ is closed. This question is étale local on Z hence we may assume Z is affine (some details omitted). Since Y is a scheme, Z is affine, and $Z \to Y$ is locally of finite presentation we can find an immersion $Z \to \mathbf{A}^n \times Y$, see Morphisms, Lemma 04II. Consider the cartesian diagram

$$\begin{array}{ccc} X \times_Y Z & \longrightarrow & \mathbf{A}^n \times X \\ \downarrow & & \downarrow \\ Z & \longrightarrow & \mathbf{A}^n \times Y \end{array} \qquad \text{inducing the cartesian square} \qquad \begin{array}{ccc} |X \times_Y Z| & \longrightarrow & |\mathbf{A}^n \times X| \\ \downarrow & & \downarrow \\ |Z| & \longrightarrow & |\mathbf{A}^n \times Y| \end{array}$$

of topological spaces whose horizontal arrows are homeomorphisms onto locally closed subsets (Properties of Spaces, Lemma 0ABJ). Thus every closed subset T of $|X \times_Y Z|$ is the pullback of a closed subset T' of $|\mathbf{A}^n \times Y|$. Since the assumption is that the image of T' in $|\mathbf{A}^n \times X|$ is closed we conclude that the image of T in $|Z|$ is closed as desired. □